

XIAOYING SONG

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RESEARCH INTERESTS

Areas: Human-Agent Collaboration · Personalized Alignment · Computational Social Science · NLP

My research develops large language model systems that collaborate with people and other agents to deliver responses aligned with diverse, evolving user needs. I work along three connected directions: (i) **human-agent collaboration**: multi-turn conversation optimization and multi-agent frameworks that clarify user intent and coordinate specialized agents [COLM 2025; EMNLP 2025]; (ii) **personalized alignment**: user-centered dialogue systems and personalized safety alignment that tailor responses to individual literacy levels, preferences, and safety needs [EMNLP 2025; arXiv 2026]; and (iii) **computational social science**: applying these methods to high-stakes problems including health misinformation, hate speech, and crisis communication [EMNLP 2024/2025; NAACL 2025]. My work has appeared at EMNLP, NAACL, COLM, and COLING, with several first-authored papers.

ACADEMIC EXPERIENCE

University of North Texas, TX, USA

Aug 2023 – Present

Ph.D. Candidate, Data Science (expected 2027); Research Assistant

Advisor: Prof. Lingzi Hong

Central China Normal University, China

Aug 2021 – Jun 2023

M.S., Information Science; Research Assistant

PUBLICATIONS

Conference Papers

1. **Song, X.**, Anik, A. S., Liu, J., Tan, Q., et al. Ask or answer: A decision framework for multi-turn health misinformation intervention. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
2. Tan, Q., **Song, X.**, Cheng, N., Li, G., et al. End-to-end on-device quantization-aware training for LLMs at inference cost. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
3. Tan, Q., **Song, X.**, Cheng, N., Liu, N., et al. Q-Realign: Piggybacking realignment on quantization for safe and efficient LLM deployment. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
4. **Song, X.**, Anik, A. S., Barua, D., Luo, P., et al. Speaking at the right level: Literacy-controlled counterspeech generation with RAG-RL. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
5. **Song, X.**, Anik, A. S., Blanco, E., et al. A dynamic fusion model for consistent crisis response. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
6. **Song, X.**, Perez, S. L., Yu, X., et al. Echoes of discord: Forecasting hater reactions to counterspeech. In *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2025.
7. **Song, X.**, Mamidisetty, S., Blanco, E., & Hong, L. Assessing the human likeness of AI-generated counterspeech. In *International Conference on Computational Linguistics (COLING)*, 2025.
8. **Song, X.**, Anik, A. S., Frías-Martínez, V., & Hong, L. Dynamic fusion of large language models for crisis communication. In *International Conference on Information Systems for Crisis Response and Management (ISCRAM)*, 2025.
9. **Song, X.**, Anik, A. S., Wang, E., Wang, B., et al. Multi-agent retrieval-augmented framework for evidence-based counterspeech against health misinformation. In *Conference on Language Modeling (COLM)*, 2025.
10. Hong, L., Luo, P., Blanco, E., & **Song, X.** Outcome-constrained large language models for countering hate speech. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.
11. Liu, J., **Song, X.**, Zhang, D., Thomale, J., et al. A hybrid framework for subject analysis: Integrating embedding-based regression models with large language models. In *Annual Meeting of the Association for Information Science and Technology (ASIS&T)*, 2025.
12. Tan, Q., **Song, X.**, et al. Palette: A modular, controllable, and efficient framework for on-demand authorized safety alignment relaxation in LLMs. *arXiv:2605.24154*. Under review at **NeurIPS** 2026.
13. **Song, X.**, et al. ST-Crisis: A diagnostic benchmark for spatiotemporal grounding in crisis communication. Under review at **EACL** 2026.

14. **Song, X.**, et al. Towards human-centered natural language processing: A survey of cognitive signal integration. Under review at **EACL** 2026.
15. Anik, A. S., **Song, X.**, Kanuganti, S., Yuan, J., et al. Granularity-aligned health misinformation refutation with knowledge graphs. Under review at **NAACL** 2026.

Journal Papers

1. **Song, X.**, Luo, P., Thomale, J., et al. Comparative analysis of large language models' performance in book classification tasks using Library of Congress Classification system. *Journal of Information Science*, 2026.
2. Perez, S. L., **Song, X.**, & Hong, L. Analyzing the language of rejection: A study of user flagging responses to hate speech on Reddit. *Information Research*, 2025.
3. Tan, Q., **Song, X.**, Ye, G., et al. An effective negative sampling approach for contrastive learning of sentence embedding. *Machine Learning*, 2023.

PROJECT EXPERIENCE

User-Centered Multi-Turn Conversation Optimization Project Leader · Jan 2026 – Present

Led the design of **RO-PnR**, a reinforcement-learning framework that decides *whether to ask a clarifying question or answer directly*, weighing a probe's expected information gain against its cost. Achieved the highest cost-adjusted utility with ~30% fewer turns than always-probe baselines. **Outcomes:** 1 EMNLP 2026 paper, 1 in progress.

Knowledge-Enhanced Crisis Response and Misinformation Mitigation Project Leader · Aug 2024 – Present

Led two retrieval-augmented systems: a consistency-optimized fusion model for crisis response (0.81–0.85 quality across 3 base models, beating single-model baselines) and a multi-agent, literacy-controlled counterspeech framework for health misinformation. **Outcomes:** 3 papers published (2 EMNLP 2025, 1 COLM 2025), 2 in progress.

Spatiotemporal Reasoning in Large Language Models Project Leader · Aug 2025 – Present

Built **ST-Crisis**, a diagnostic benchmark and RAG framework measuring spatiotemporal grounding in crisis response; diagnosed 5 LLMs across three model families, exposing where they fail to ground evidence in location and time. **Outcomes:** 2 under review, 1 in progress.

These projects are supported by IMLS National Leadership Grant LG-256661-OLS-24 and NSF Award 2225229.

HONORS AND AWARDS

Second Place , CLEF 2026 CheckThat! Lab Competition	2026
Honorable Mention , ASIS&T SIG-SM Student Competition	2026
First Place , UNT Phil and Lis Turner Outstanding Paper Award	2025
UNT Porter-Evans Scholarship	2025
UNT COI/IS Leon M. Liddell Scholarship	2024
UNT Graduate School Travel Award	2024
CCNU Excellent Student Awards	2023
CCNU Graduate Scholarships	2023

INVITED TALKS, TEACHING & MENTORSHIP

Invited Tutorials IDEA Institute on AI (ASIS&T), with Prof. Lingzi Hong

- *AI-assisted Communications* 2025, 2026
- *Data Collection, Classification, and Transformation* 2024
- *Comprehending the Semantic Essence of Text: Leveraging SEANCE for Textual Analysis* 2024

Teaching Assistant UNT, Aug 2023 – Jul 2024

Introduction to Data Science; Data Visualization; Information Architecture

Mentoring & Advising Jan 2024 – Present

- Mentor, UNT AI Summer Program – guided master's and undergraduate students through AI research projects; mentees co-authored papers at COLM and EMNLP 2024, 2025, 2026
- Advised Sujana Mamidisetty, Diana Zhang, and Elliott Wang – admitted to computer science programs at UMD, UT, and Purdue 2023 – 2026

TECHNICAL SKILLS & SERVICE

LLM Training & Alignment: Supervised fine-tuning and RL-based alignment (PPO, GRPO) with PEFT/LoRA.

LLM Systems & Evaluation: Retrieval-augmented generation, multi-agent frameworks, synthetic dataset and benchmark construction, LLM-as-judge evaluation validated against human annotation.

Academic Service: Reviewer for ACL, NAACL, EMNLP, COLING, COLM, NeurIPS, JCDL, and ASIS&T.